

OWASP Juice Shop Vulnerability Assessment Report

Student Name: Praveen Podili

Application Tested: OWASP Juice Shop

Testing Date: 24 July 2026

Objective

The objective of this assessment was to identify and demonstrate web application vulnerabilities using the OWASP Juice Shop training application. The testing was performed in a controlled environment for educational purposes.

Environment

Target Application

- OWASP Juice Shop

Browser

- FireFox, Google Chrome

Operating System

- Linux , Windows

Vulnerability 1 – SQL Injection

Description

SQL Injection occurs when user-supplied input is interpreted as SQL commands due to improper input validation.

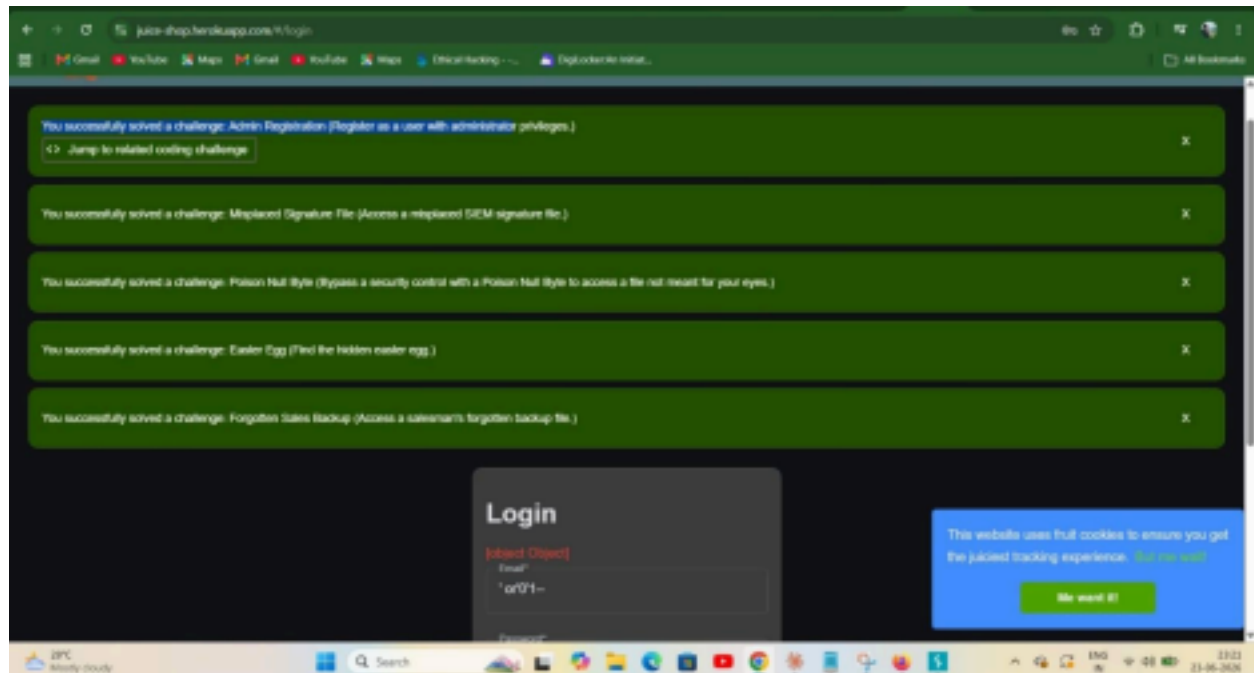
Evidence

On the login page, the following SQL Injection payload was entered into the Email field:

```
' or '0'='1--
```

This demonstrates testing of authentication input for SQL Injection.

Screenshot



OWASP Category

A03:2021 – Injection

Recommendation

- Use parameterized SQL queries.
- Validate all user inputs.
- Escape special SQL characters.
- Avoid dynamic SQL queries.

Vulnerability 2 – Admin Registration

Description

The "Admin Registration" challenge demonstrates a Broken Access Control vulnerability by allowing registration of a user with administrator privileges.

Status

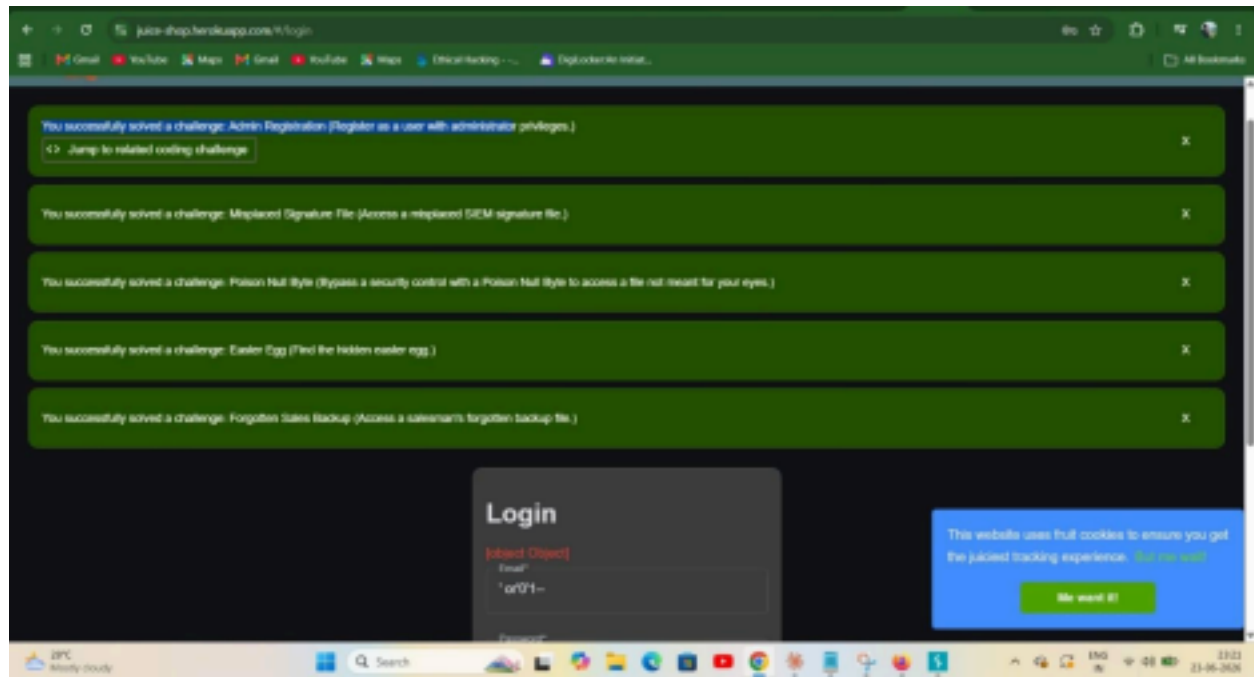
✓ Successfully Completed

Evidence

Visible in the green challenge notification:

Successfully solved a challenge: **Admin Registration**

Screenshot



OWASP Category

A01:2021 – Broken Access Control

Recommendation

- Restrict administrator registration.
- Perform server-side role validation.
- Prevent privilege escalation.

Vulnerability 3 – Misplaced Signature File

Description

The "Misplaced Signature File" challenge demonstrates Information Disclosure by exposing a file that should not be publicly accessible.

Status

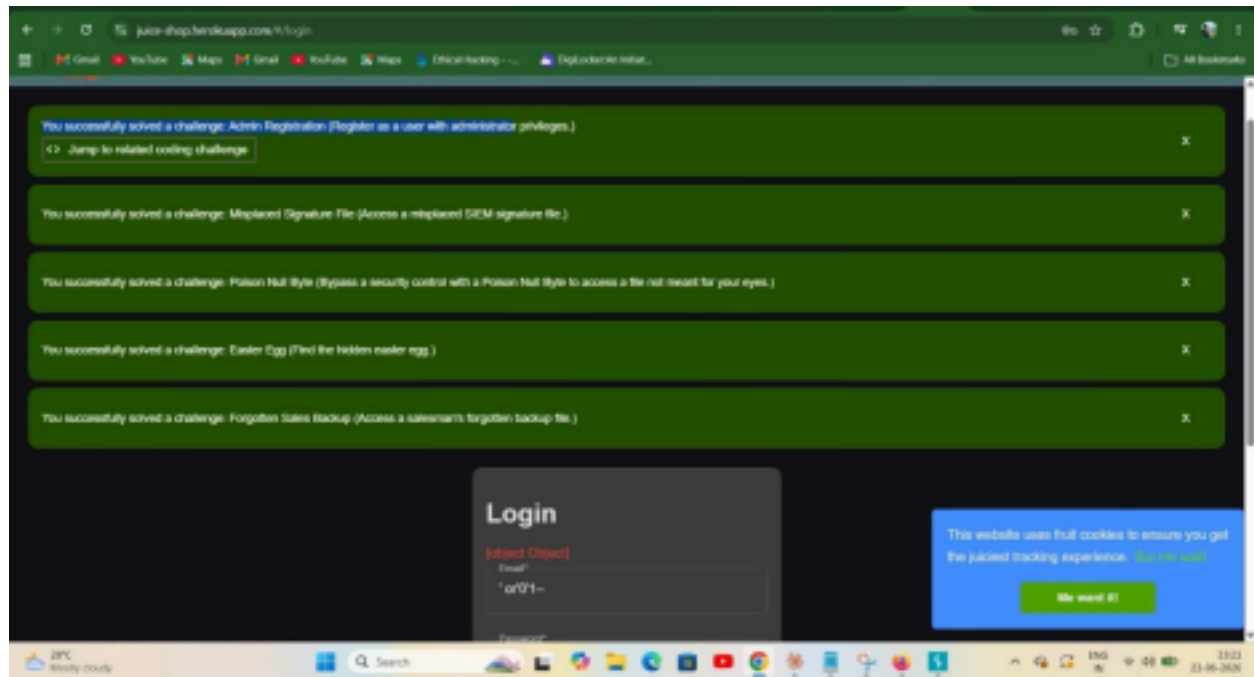
✓ Successfully Completed

Evidence

Visible in the notification:

Successfully solved a challenge: **Misplaced Signature File**

Screenshot



OWASP Category

A05:2021 – Security Misconfiguration

Recommendation

- Remove unnecessary files.
- Restrict access to sensitive resources.
- Review deployment configurations.

Vulnerability 4 – Poison Null Byte

Description

The "Poison Null Byte" challenge demonstrates improper file validation that may allow unauthorized access through crafted input.

Status

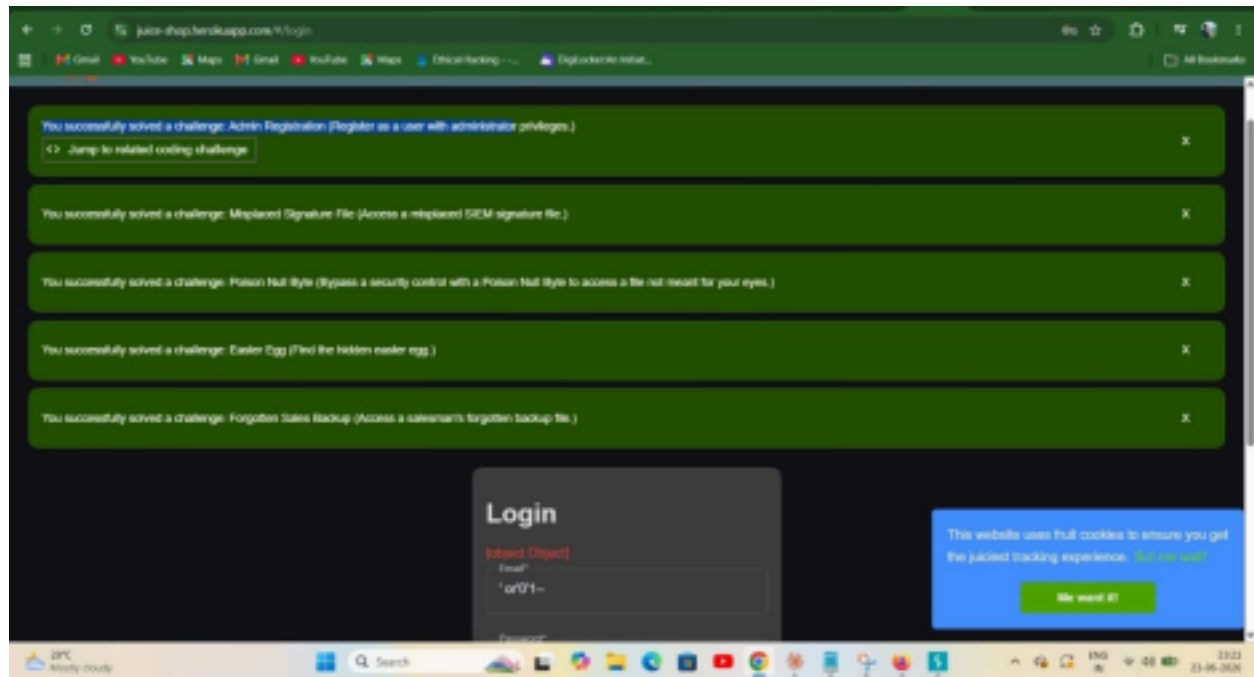
✓ Successfully Completed

Evidence

Visible in the notification:

Successfully solved a challenge: **Poison Null Byte**

Screenshot



OWASP Category

A01:2021 – Broken Access Control

Recommendation

- Sanitize file paths.
- Validate filenames.
- Prevent directory traversal.
- Reject null-byte injection attempts.

Vulnerability 5 – Forgotten Sales Backup

Description

The "Forgotten Sales Backup" challenge demonstrates Sensitive Information Disclosure caused by exposed backup files.

Status

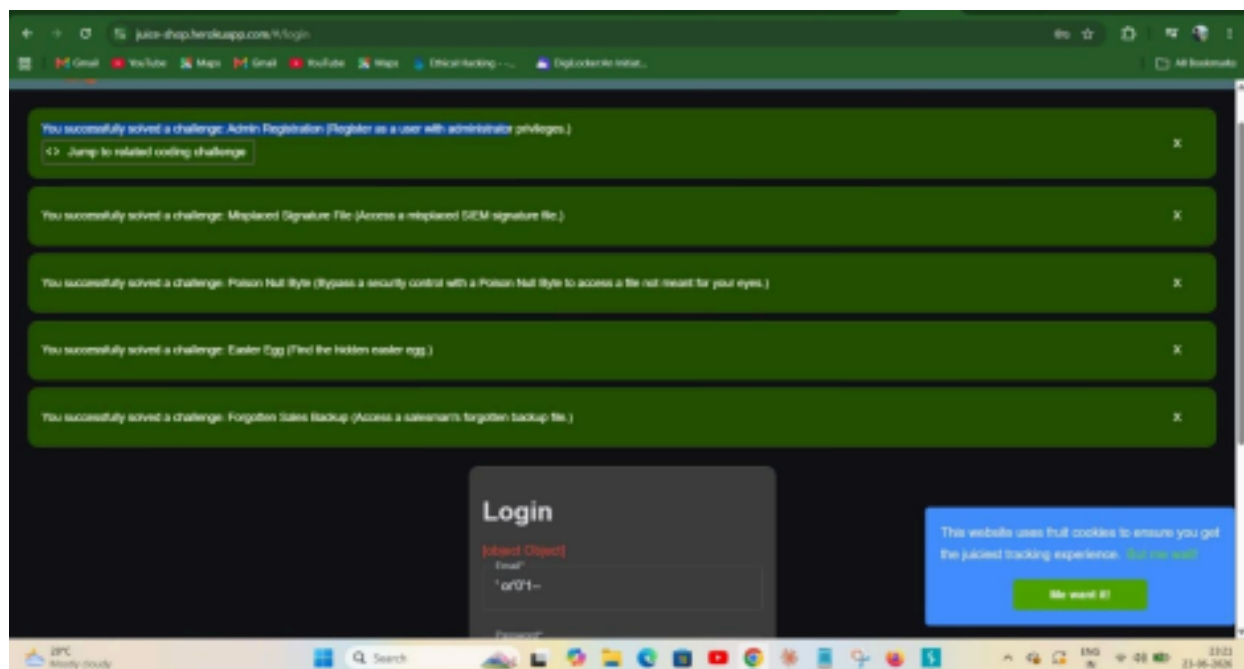
✓ Successfully Completed

Evidence

Visible in the notification:

Successfully solved a challenge: **Forgotten Sales Backup**

Screenshot



OWASP Category

A05:2021 – Security Misconfiguration

Recommendation

- Remove backup files from production.
- Protect sensitive resources.
- Perform regular security reviews.

Additional Challenge

Easter Egg

The hidden Easter Egg challenge was also successfully completed.

Although this is primarily a training challenge rather than a vulnerability itself, it demonstrates discovery of hidden application resources.

Summary

Challenge	Vulnerability	Category Status
SQL Injection	Injection	✓ Completed
Admin Registration	Broken access control	✓ Completed
Misplaced Signature File	Information Disclosure	✓ Completed
Poison Null Byte	File Validation / Access Control	✓ Completed
Forgotten Sales Backup	Sensitive Information Disclosure	✓ Completed
Easter Egg	Hidden Resource Discovery	✓ Completed

Conclusion

The OWASP Juice Shop application was assessed for multiple security challenges. Evidence from the testing confirms successful completion of SQL Injection testing and several Juice Shop challenges related to broken access control, information disclosure, file exposure, and security misconfiguration. These findings demonstrate common web application security weaknesses and highlight the importance of secure input validation, proper authorization checks, and protection of sensitive resources.